EX-14. Network traffic analysing using wireshark

Aim:

To capture and analyze network traffic using wireshark.

Procedure:

1. Install and open wireshark.
2. Select the active network interface.
3. Start capturing packets.
4. Generate network traffic.
5. Apply suitable display filters
6. Analyze TCP, UDP, IP, ARP, DHCP, ICMP, and DNS packets.
7. Stop and save the packets captured.

Result:

Thus, the network packets were captured and analyzed successfully using wireshark.

EX-14.

wlp0s20f3

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	18.97.36.57	172.16.73.138	TLSv1.2	184	Application Data
2	0.000070605	172.16.73.138	18.97.36.57	TCP	66	52006 → 443 [ACK] Seq=1…
3	0.100252336	fe80::a430:edff:fe5…	ff02::fb	MDNS	506	Standard query response…
4	0.100336704	172.16.75.1	224.0.0.251	MDNS	82	Standard query 0x0000 P…
5	0.100337550	172.16.75.1	224.0.0.251	MDNS	82	Standard query 0x0000 P…
6	0.404914249	172.16.79.102	224.0.0.251	MDNS	151	Standard query response…
7	0.709760937	AzureWaveTec_d5:74:…	Broadcast	LLC	60	S P, func=RNR, N(R)=64;
8	0.709773365	172.16.72.175	255.255.255.255	UDP	763	54187 → 29810 Len=721
9	2.183123892	172.16.73.138	160.79.104.10	TLSv1.2	2843	Application Data
10	2.191828227	172.16.73.138	160.79.104.10	TLSv1.2	1819	Application Data
11	2.212301723	160.79.104.10	172.16.73.138	TCP	66	443 → 59038 [ACK] Seq=1…
12	2.258463969	172.16.75.176	224.0.0.251	MDNS	77	Standard query 0x0000 P…
13	2.258477075	172.16.78.153	224.0.0.251	MDNS	428	Standard query response…
14	2.278670963	160.79.104.10	172.16.73.138	TCP	66	443 → 59038 [ACK] Seq=1…
15	2.278671396	160.79.104.10	172.16.73.138	TCP	66	443 → 59038 [ACK] Seq=1…
16	2.352652912	172.16.73.138	57.144.57.32	TCP	145	46506 → 5222 [PSH, ACK]…
17	2.365516669	57.144.57.32	172.16.73.138	TCP	66	5222 → 46506 [ACK] Seq=…
18	2.653708230	160.79.104.10	172.16.73.138	TLSv1.2	221	Application Data
19	2.653708725	160.79.104.10	172.16.73.138	TLSv1.2	140	Application Data
20	2.653708789	160.79.104.10	172.16.73.138	TLSv1.2	97	Application Data
21	2.653708851	57.144.57.32	172.16.73.138	TCP	147	5222 → 46506 [PSH, ACK]…
22	2.653782645	172.16.73.138	160.79.104.10	TCP	66	59038 → 443 [ACK] Seq=4…
23	2.653811365	172.16.73.138	160.79.104.10	TCP	66	59038 → 443 [ACK] Seq=4…
24	2.653827338	172.16.73.138	57.144.57.32	TCP	66	46506 → 5222 [ACK] Seq=…
25	2.653853732	172.16.73.138	160.79.104.10	TCP	66	59038 → 443 [ACK] Seq=4…

```
▾ Frame 2: Packet, 66 bytes on wire (528 bits), 66 bytes captured (
    Section number: 1
  ▸ Interface id: 0 (wlp0s20f3)
    Encapsulation type: Ethernet (1)
    Arrival Time: Oct 14, 2026 08:59:11.405691957 IST
    UTC Arrival Time: Oct 14, 2026 03:29:11.405691957 UTC
    Epoch Arrival Time: 1791948551.405691957
    [Time shift for this packet: 0.000000000 seconds]
    [Time delta from previous captured frame: 70.605 microseconds]
    [Time delta from previous displayed frame: 70.605 microseconds]
    [Time since reference or first frame: 70.605 microseconds]
    Frame Number: 2
    Frame Length: 66 bytes (528 bits)
    Capture Length: 66 bytes (528 bits)
    [Frame is marked: False]
    [Frame is ignored: False]
    [Protocols in frame: eth:ethertype:ip:tcp]
    Character encoding: ASCII (0)
    [Coloring Rule Name: TCP]
    [Coloring Rule String: tcp]
▸ Ethernet II, Src: Intel_ca:3b:e5 (44:a3:bb:ca:3b:e5), Dst: Sophos
▸ Internet Protocol Version 4, Src: 172.16.73.138, Dst: 18.97.36.57
▸ Transmission Control Protocol, Src Port: 52006, Dst Port: 443, Se
```

```
0000  7c 5a 1c d2 93 f1 44 a3  bb ca 3
0010  00 34 81 2b 40 00 40 06  8d 64 a
0020  24 39 cb 26 01 bb c3 7b  9f 6a 3
0030  00 3b 2c 5b 00 00 01 01  08 0a (
0040  97 22
```

wireshark_wlp0s20f3MT40W3.pcapng

Packets: 114 · Dropped: 0 (0.0%)

Profile: Default